JIAYE CHEN, Ph.D.

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EDUCATION & APPOINTMENTS

May 2026 – Present | Postdoctoral Research Fellow, Molecular Foundry, Lawrence Berkeley National Laboratory. Supervisors: Dr. Emory Chan and Dr. Bruce Cohen

March 2025 – April 2026 | Postdoctoral Research Fellow, Department of Chemistry, National University of Singapore. Supervisor: Prof. Xiaogang Liu.

August 2020 – December 2024 | Ph.D. in Chemistry, National University of Singapore. Supervisors: Prof. Xiaogang Liu and Prof. Qian He. Dissertation: *Manipulating Upconversion Properties through Crystal Engineering in Lanthanide-Doped Nanocrystals*.

July 2019 – September 2019 | Visiting student, Department of Chemical & Environmental Engineering, Yale University. Supervisor: Prof. Shu Hu.

September 2016 – June 2020 | B.Sc. in Materials Chemistry, University of Science and Technology of China.

RESEARCH DIRECTIONS

- **Photon Upconversion and Nonlinear Photophysics:** investigating photon upconversion, energy transfer mechanisms, and ultrahigh optical nonlinearities through single-particle imaging and time-resolved photophysical analysis, aiming to unravel fundamental principles for super-resolution imaging, ultrasensitive sensing, and quantum photonics.
- **Nanomaterials Synthesis and Structural Characterization:** designing and synthesizing functional nanostructures using both conventional manual methods and automated high-throughput nanocrystal synthesis platforms, while applying advanced electron microscopy and synchrotron spectroscopy to elucidate synthesis–structure–property relationships.
- **AI-Enabled Materials Discovery:** developing domain-grounded LLM agents, machine-learning models, and rate-equation frameworks to understand lanthanide energy transfer and accelerate materials discovery.

TECHNICAL EXPERTISE

- **Synthesis & Fabrication:** lanthanide-doped nanocrystals, core–multishell architectures, sublattice/lattice reconstruction and vacancy engineering, automated high-throughput synthesis and thin-film/protective-coating deposition.
- **Optical Spectroscopy & Imaging:** single-particle spectroscopy, time-resolved photoluminescence, photon-upconversion and optical-nonlinearity measurements, and super-resolution optical imaging.
- **Structural Characterization:** scanning/high-resolution transmission electron microscopy and synchrotron X-ray absorption spectroscopy.
- **Computation & AI for Science:** structural modeling, machine learning, and multi-agent LLM pipelines for scientific data extraction; Python, MATLAB, and standard analysis/visualization tools (Artemis/Athena, DigitalMicrograph, CrystalMaker, VESTA, ImageJ, OriginLab, 3ds Max, Adobe Illustrator).

CORE CONTRIBUTIONS

- **Sub-10-nm bright upconversion nanocrystals via dynamic metal-complex docking:** achieved a two- to three-order-of-magnitude luminescence enhancement through volumetric vacancy removal, enabling bright probes for cell labeling and neuron tracking, and providing fundamental insights into vacancy diffusion.
- **27-nm sublattice-reconstructed nanocrystals with nonlinearity of 156:** Lu^{3+}-induced crystal-field distortion amplified cross-relaxation and accelerated avalanche response to 9 ms (70-fold faster than previous systems), enabling 33-nm lateral (~λ/32) and 80-nm axial (~λ/13) single-beam super-resolution imaging as a low-cost alternative to STED and STORM.
- **176-nm nanodiscs with optical nonlinearity exceeding 500:** delivered a milestone advance in lanthanide-based nonlinear nanophotonics, surpassing the long-standing nonlinearity ceiling of 60. Regional differentiation within single nanodiscs enabled imaging below the physical probe size, revealing spatially resolved energy-transfer heterogeneity and offering a new materials platform for next-generation nanoprobes, optical computing, and integrated photonic chips.
- **AI-ready scientific knowledge infrastructure:** developed domain-grounded multi-agent LLM pipelines that convert unstructured scientific PDFs into entity-resolved knowledge repositories across text, tables, and figures. Integrated pixel-level reconstruction of figure-embedded spectral data to enable high-fidelity literature-to-data distillation, materials discovery, and cross-domain scientific reasoning.

PUBLICATIONS

1. **Chen, J.**, Liu, C., Xi, S., Tan, S., He, Q., Liang, L., & Liu, X. (2025). Optical nonlinearities in excess of 500 through sublattice reconstruction. *Nature* 643, 669-674.
    - *Highlighted* in: *Nature Physics* 21, 1186 (2025); *Light: Science & Applications* 14, 281 (2025); *Chinese Journal of Structural Chemistry* 44, 100682 (2025).
    - *Featured* by NUS News, EurekAlert!, Phys.org, Photonics Online, China Science Daily, Guangming Online, ScienceNet.cn, etc. (June 2025).
2. **Chen, J.**, & Liu, X. (2025). Unlocking an optical bistability switch. *Nature Photonics* 19, 122-123.
3. **Chen, J.**, Liang, L., Tan, S., Xi, S., Lin, C. H., Wu, T., He, Q., & Liu, X. (2023). Volumetric nanocrystal lattice reconstruction through dynamic metal complex docking. *Nano Letters* 23, 7221-7227.
4. Zhang, X.†, **Chen, J.**†, Wang, C. W.†, Liang, L., Dai, M., Zhang, Y., Liu, X., & Wang, Q. J. (2026). Visible-readout upconversion nanotransducers for mid-infrared photodetection. *Advanced Functional Materials* 36, 76826.
5. Zhang, X.†, Wang, C. W.†, **Chen, J.**†, Dai, M., Luo, Y., Liang, L., Liu, X., & Wang, Q. J. (2026). Plasmonic-enhanced lanthanide nanotransducers for mid-infrared detection. *Nature Communications* 17, 10305.
6. Wang, C. W., Liang, L., Zhang, X., Dai, M., **Chen, J.**, Liu, X., & Wang, Q. J. (2026). Mid-infrared detection through ligand-driven local heating in lanthanide-doped nanoparticles. *Nature Communications* 17, 4306.
7. Liu, Y., **Chen, J.**, & Liu, X. (2026). Photophysics-guided upconversion nanosystems for sensing. *Accounts of Chemical Research* 59, 1043-1055.
8. Guo, D., Pang, X., Liu, C., Li, Y., **Chen, J.**, ... & Liu, X. (2025). On-demand control of lanthanide optical dynamics via pumping-flux modulation. *Nano Letters* 25, 9787-9793.
9. Schiattarella, C., Romano, S., Sirleto, L., Mocella, V., Rendina, I., Lanzio, V., ... **Chen, J.**, ... & Zito, G. (2024). Directive giant upconversion by supercritical bound states in the continuum. *Nature* 626, 765-771.
10. He, C., Qiu, J., Mu, Z., **Chen, J.**, ... & Liu, X. (2024). Room temperature circularly polarized emission in perovskite nanocrystals through bichiral-molecule-induced lattice reconstruction. *Matter* 7, 475-484.
11. Zhou, L., Liang, L., **Chen, J.**, Zhou, X., Liu, L., Xi, S., ... & Liu, X. (2024). Promoted growth and multiband emission in heterostructured perovskites through Cs^+-sublattice interaction. *Advanced Science* 11, 2306398.
12. Qiu, J., Zhao, H., Mu, Z., **Chen, J.**, Gu, H., Gu, C., Xing, G., Qin, X., & Liu, X. (2024). Turning nonemissive $CsPb_2Br_5$ crystals into high-performance scintillators through alkali metal doping. *Nano Letters* 24, 2503-2510.
13. Liang, L., **Chen, J.**, Shao, K., Qin, X., Pan, Z., & Liu, X. (2023). Controlling persistent luminescence in nanocrystalline phosphors. *Nature Materials* 22, 289-304.
14. Liang, L., **Chen, J.**, & Liu, X. (2023). Lanthanide-doped upconversion nanomaterials. *Comprehensive Inorganic Chemistry III*, 439-485.
15. He, R., Rasmita, A., Zhou, L., Liang, L., Cai, X., **Chen, J.**, ... Gao, W., & Liu, X. (2023). Magnetically tunable spontaneous superradiance from mesoscopic perovskite emitter clusters. *Journal of Physical Chemistry Letters* 14, 2627-2634.
16. Liang, L., Wang, C., **Chen, J.**, Wang, Q. J., & Liu, X. (2022). Incoherent broadband mid-infrared detection with lanthanide nanotransducers. *Nature Photonics* 16, 712-717.
17. Shen, X., Zhao, T., Su, H., Yang, M., **Chen, J.**, ... & Hu, S. (2022). Tuning intermediate bands of protective coatings to reach the bulk-recombination limit of stable water-oxidation GaP photoanodes. *Advanced Energy Materials* 12, 2201314.
18. Jin, H., Wang, H., Qi, Z., Bin, D. S., Zhang, T., Wan, Y., **Chen, J.**, ... Ji, H., & Wan, L. J. (2020). A black phosphorus-graphite composite anode for Li-/Na-/K-ion batteries. *Angewandte Chemie International Edition* 59, 2318-2322.

AWARDS & EXTERNAL RECOGNITION

April 2026 | Singapore NRF Postdoctoral Award, Singapore National Research Foundation (competitive national fellowship with a total award value of S$650,000, including an independent S$250,000 research grant)

June 2024 | IUMRS-ICYRAM Young Researcher Award, International Union of Materials Research Societies (20 awardees worldwide per year)

August 2023 | NUS Research Incentive Award, National University of Singapore (top-up award for research excellence beyond the President's Graduate Fellowship)

July 2020 | President's Graduate Fellowship, National University of Singapore (the university's most prestigious graduate fellowship)

May 2020 | Outstanding Graduate, Anhui Province, China (top 3% of undergraduate graduates in Anhui Province)

May 2020 | Outstanding Graduate, University of Science and Technology of China (top 15% of graduating class)

SELECTED INVITED PRESENTATIONS

October 2025 | Optical Nonlinearities in Excess of 500 through Sublattice Reconstruction, SHIFT 2025, University of La Laguna, Tenerife, Spain.

June 2024 | Advancing Optical Applications through Crystal Engineering and Energy Manipulation in Lanthanide-Doped Nanomaterials, award lecture at ICYRAM Young Researcher Symposium, hosted by the International Union of Materials Research Societies, Jeju Island, South Korea.

TEACHING, MENTORING & SERVICE

Teaching

- Teaching Assistant and Lab Instructor, CM3292, National University of Singapore (2020–2023), supervised undergraduate laboratory sessions, led tutorials and discussion sections, and guided experimental techniques and lab-report assessment.

Mentoring

- Mentored 11 junior researchers in nanocrystal synthesis, single-particle spectroscopy, electron microscopy, and large language models, including 5 Ph.D., 4 master's, and 2 undergraduate students at NUS and LBNL.
- Onboarded and trained new lab members and junior postdocs on instrumentation and experimental workflows.

Professional Service

- Led inorganic sub-group journal-club and group-meeting discussions on research progress and literature.
- Built lab documentation and knowledge-base resources.
- Science outreach and public engagement, including introducing the super-resolution microscope and WANDA robotic nanocrystal-synthesis platform.